**ASSIGNMENT-2**

1. Design a complete class hierarchy for a banking system incorporating:
    - inheritance
    - polymorphism
    - encapsulation
    - static variables

    Explain how each concept contributes to system design.
2. Critically compare:
    - pure Python data structures
    - NumPy arrays
    - Pandas DataFrames
    - in terms of memory model, computational efficiency, scalability, and usability. Support your answer with scenarios where each approach is preferable.
3. Design a hierarchy of custom exceptions for a **file validation system** (e.g., invalid format, missing data, and permission issues). Explain how this improves debugging and system clarity.